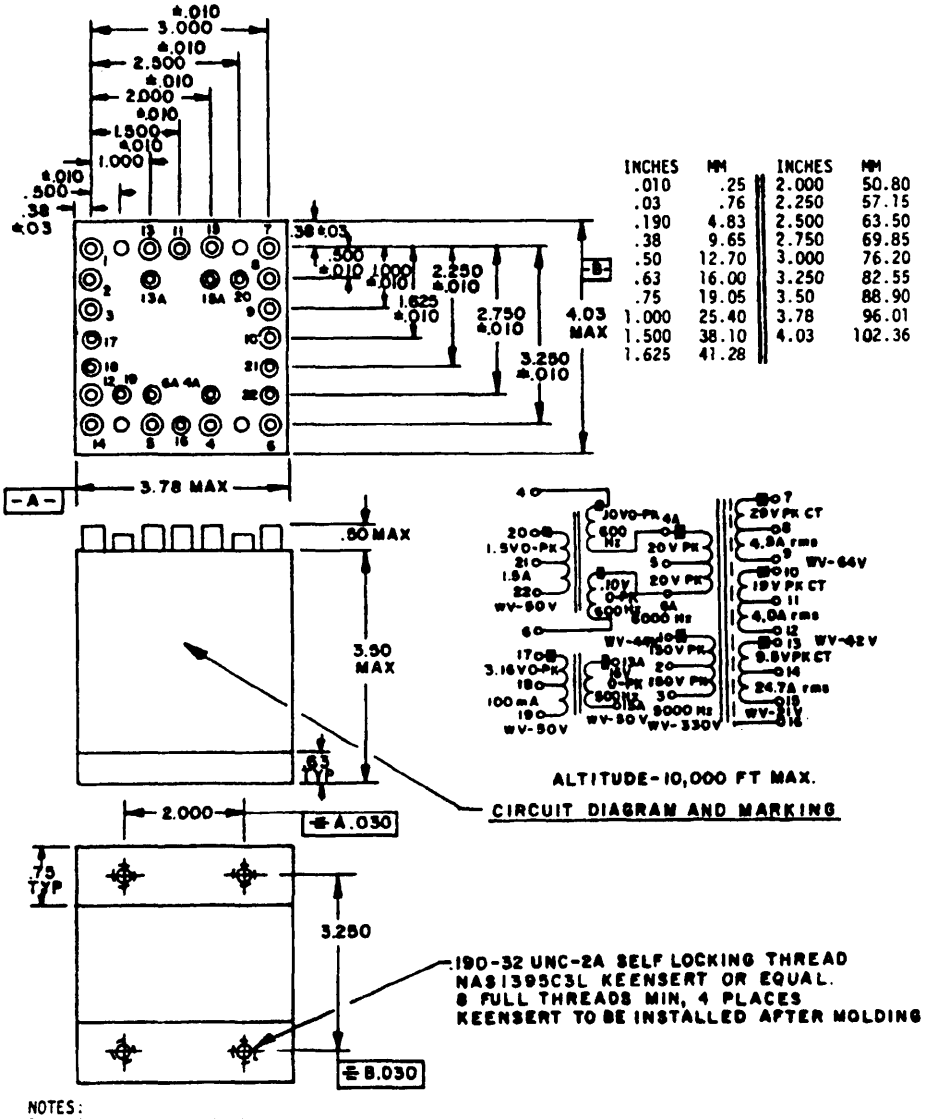
MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, TF5S03ZZ940

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of of this document and the latest issue of Specification MIL-T-27.



INCHES	MM	INCHES	MM
.010	.25	2.000	50.80
.03	.76	2.250	57.15
.190	4.83	2.500	63.50
.38	9.65	2.750	69.85
.50	12.70	3.000	76.20
.63	16.00	3.250	82.55
.75	19.05	3.50	88.90
1.000	25.40	3.78	96.01
1.500	38.10	4.03	102.36
1.625	41.28		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Marking shall be on the side of the case.
4. Tolerance on terminal position dimensions, non-cumulative.

FIGURE 1. Dimensions and configuration.

FSC 5950

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-3) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:
  Transformer 1:
    Primary 1 (1-2-3): 150 V peak square wave applied to half primary alternately, 5,000 Hz or
    Primary 2 (4A-5-6A): 20 V peak square wave applied to half primary alternately, 6,000 Hz.
    Secondary 1 (7-8-9): 29 V peak line to center-tap, 4.5 amperes rms line-to-line. (This secondary supplies +28 V dc at 6 amperes dc and -28 V dc at 2 amperes dc from two full wave center-tapped rectifiers).
    Secondary 2 (10-11-12): 19 V peak line to center-tap, 4 amperes rms line-to-line. (This secondary supplies +18 V dc at 4 ampere dc and -18 V dc at 4 amperes dc from two full wave center-tapped rectifiers).
    Secondary 3 (13-14-15): 9.5 volts peak line to center-tap, 24.7 amperes rms line-to-line. (This secondary supplies +8 V dc at 35 amperes dc from a full wave center-tapped rectifier).
  Transformer 2:
    Primary 1: (4-4A): } 0.1 volt, 0 to peak, 600 Hz square wave alternately.
    Primary 2: (6-6A): }
    Secondary (20-21-22): 1.5 volts, 0 to peak, 1.5 amperes, 0 to peak on each half alternately.
  Transformer 3:
    Primary (13A-15A): 16 volts, 0 to peak, 5,000 Hz square wave.
    Secondary (17-18-19): 3.16 volts, 0 to peak, 100 mA, 0 to peak, one-half secondary.

Working voltage:
  (1-3): 330 volts
  (4A-6A): 44 volts
  (7-9): 64 volts
  (10-12): 42 volts
  (13-15): 21 volts
  (4-4A); (6-6A), (13A-15A), (17-19), and (20-22): 50 volts, respectively.

Design and construction:
  Dimensions and configuration: See figure.
  Duty cycle: Continuous.
  Case: Molded.
    Material: Alumina-filled epoxy.
  Altitude: 10,000 feet maximum.
  Terminals: 11 Cambion No. 1021-3 or equal (gold-plated), and 15 Cambion No. 1023-5 or equal (gold-plated).

Terminal height: 0.5 inch maximum.

Weight: 8 pounds, maximum.

Operating temperature range: -55° to +130°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 5 pounds; and test condition E, 5 pound-inches for 1021-3 and 15 pound-inches for 1023-5.

Dielectric withstanding voltage (at sea level):
  Transformer 1:
    (1-3) and (16): 1,000 volts rms, respectively.
    (4A-6A), (10-12), and (13-15): 100 volts rms, respectively.
    (7-9): 300 volts rms.

  Transformer 2: (20-22) 100 volts rms

  Transformer 3: (13A-15A) and (17-19) 100 volts rms, respectively.

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Electrical characteristics:

No load: Center tap accuracy for transformers 1, 2, and 3 shall be ±0.5 percent.

Transformer 1:

(a) With 167 V rms and 5,000 Hz across (1-2), current in (1-2) shall not exceed 0.2 ampere and power in (1-2) shall not exceed 18 watts.

(b) With 150 V rms, 1,500 Hz across (1-2), voltage across:
(4A-6A) and (10-12) shall be 40.54 ±1 percent, respectively.
(7-9) shall be 60.81 V ±1 percent.
(13-15) shall be 20.27 V ±1 percent.

Transformer 2: With 3.0 volts rms, 6,000 Hz across (20-22), voltage across (4-4A) and (6-6A) shall be 0.1 V ±1 percent, respectively.

Transformer 3: With 8.8 volts rms, 5,000 Hz across (13A-15A), voltage across (17-18) and (19-18) shall be 1.6 V ±1 percent, respectively.

Rated load: With the rated primary voltage and current at the rated frequency across the primary, the secondary voltages shall equal the rated secondary voltage within ±5 percent.

Electrostatic shielding: Voltage ratio 3.5:1 minimum ratio at 20 kHz.

DC resistance:

Transformer 1:
(1-3) 0.42 ohm maximum.
(4A-6A) 0.009 ohm maximum.
(7-9) 0.054 ohm maximum.
(10-12) 0.066 ohm maximum.
(13-15) 0.0045 ohm maximum.

Transformer 2:
(4-4A) and (6-6A) 0.0017 ohms maximum, respectively.
(20-22) 0.25 ohm maximum.

Transformer 3:
(13A-15A) 6 ohms maximum.
(17-19) 1.25 ohms maximum.

Polarity:

Transformer 1:
Additive, with terminals 3 and 4A connected, 6A and 7 connected.
9 and 10 connected and with 12 and 13 connected.

Transformer 2: Additive, with terminals 4A and 6A connected and with 6 and 20 connected.

Transformer 3: Additive, with terminals 15A and 17 connected.

Saturation voltage:

Transformer 2:
2.28 volts rms ±10 percent with a 6,000 Hz sine wave voltage applied across (20-21) with all other windings open circuited.

Transformer 3:
15.6 volts rms ±10 percent with a 5,000 Hz sine wave voltage applied across (13A-15A) with all other windings open circuited.

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Temperature rise: 40°C with 167 volts rms, 5,000 Hz sine wave across (1-2) at an ambient temperature of 90°C. Secondaries loaded as follows:
(7-9) 4.5 amperes rms
(10-12) 4.0 amperes rms
(13-15) 24.7 amperes rms

Shock:(specified pulse): MIL-STD-202, method 213, test condition K, under no load conditions.

Marking location: See figure.
Part number: M27/121-01.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, SG
Navy - EC, SH, OS
Air Force - 11, 17, 99
DSA - ES

User activities:
Army - MU, WC, ME
Navy - MC, AS, SH, OS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5950-0522-5)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-T-27/121 TRANSFORMER, POWER, TYPE TF5S03Z2940

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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